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POWER SUPPLY DETECTION APPARATUS AND CABINET

Abstract

A power supply detection apparatus includes a casing, a power shelf connector, a copper bar, and a transformer. The power shelf connector is electrically connected to the power shelf. The power shelf provides a power supply to a load through the power supply detection apparatus. The copper bar is fixedly connected to the power shelf connector. The copper bar includes a first bent component and a second bent component. The first bent component and the second component are bent relative to each other and form a cavity. The transformer extends through the cavity. The transformer detects an output current and/or an output voltage of the power shelf. A cabinet is also provided, which can conveniently and quickly detect the current and the voltage outputted by the power shelf.

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Background/Summary

TECHNICAL FIELD

[0001] The present application generally relates to server cabinet technology, and particularly to a power supply detection apparatus and a cabinet.

BACKGROUND

[0002] With a development in technologies, such as big data and artificial intelligence, servers in a large-scale are widely used. The servers are usually powered by a cabinet. The cabinet includes an alternating current (AC) power supply, a power shelf, and a female cable for connecting with the servers. However, detections of a current and/or a voltage outputted by the power shelf may be problematic.

[0003] There is room for improvement in the art.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Implementations of the present application will now be described, by way of example only, with reference to the attached figures.

[0005] FIG. 1 is a diagram illustrating an embodiment of a cabinet according to the present application.

[0006] FIG. 2 is a diagram illustrating a first embodiment of a power supply detection apparatus according to the present application.

[0007] FIG. 3 is a diagram illustrating a second embodiment of a power supply detection apparatus according to the present application.

[0008] FIG. 4 is a diagram illustrating of the power supply detection apparatus of FIG. 3, viewed from another perspective according to the present application.

DETAILED DESCRIPTION

[0009] The following clearly describes the technical solution in embodiments of this application with reference to the accompanying drawings in the embodiments of the application.

[0010] It is understood that, the connection relationship described in this application is a direct connection or an indirect connection. For example, that A is connected to B may not only be that A is directly connected to B, but also be that A is indirectly connected to B by one or more other electrical components. For example, it may be that A is directly connected to C, and C is directly connected to B. In this way, A is connected to B by C. It may be further understood that “A is connected to B” described in this application may be that A is directly connected to B, or maybe that A is indirectly connected to B by one or more other electrical components.

[0011] In the description of the present disclosure, unless other specified, “/” means “or”. For example, A/B may indicate A or B. In this application, “and/or” describes only an association relationship for describing associated objects and indicates that three relationship may exist. For example, A and/or B may indicate the following three cases: Only A exists, both A and B exist, and only B exists.

[0012] In the description of the present disclosure, the words such as “first”, “second”, and the like, are used to distinguish different objects, and do not limit quantities and execution sequences. In addition, the words such as “first”, “second”, do not necessarily limit a definite difference. In addition, terms “include” and “have”, and any variant thereof are intended to cover the non-exclusive inclusion.

[0013] With a development in technologies, such as big data and artificial intelligence, servers in a large-scale are widely used. The servers are usually powered by a cabinet. The cabinet includes an

alternating current (AC) power supply, a power shelf, and a female cable for connecting with the servers. However, how to detect a current and/or a voltage outputted by the power shelf becomes a problem.

[0014] Therefore, the present application provides a power source detection apparatus and a cabinet, which is conveniently and quickly detect a current and/or a voltage outputted by the power shelf, and which has a highly integrated and is suitable for an application scenario with a small volumetric requirement.

[0015] In detail, referring to FIG. 1, FIG. 1 shows a diagram illustrating the cabinet **10** provided by the present application. The cabinet **10** includes an alternating current (AC) power supply **11**, a power shelf **12**, a power supply detection apparatus **13**, a female cable **14**, and a load **15**.

[0016] The AC power supply **11** is electrically connected to the power shelf **12**. The AC power supply is configured to output an AC to the power shelf **12**.

[0017] The power shelf **12** is electrically connected to the power supply detection apparatus **13** through the female cable **14**. The power shelf is configured to convert and output the AC outputted by the AC power supply **11**. For example, the power shelf **12** may include at least one AC/DC module, for converting the received AC into a direct current and then to output. For another example, the power shelf **12** may include at least one power distribution unit (PDU), for implementing a distribution of the received AC and a fault protection, and the like.

[0018] The power supply detection apparatus **13** is configured to detect parameters of electrical signals outputted by the power shelf **12**. For example, the power supply detection apparatus **13** may detect an output current, an output voltage, an output power of the power shelf **12**, and the like. Therefore, by detecting the parameters of the electrical signals outputted by the power shelf **12**, situations of whether a working state of the power shelf **12** is abnormal, the electrical signals outputted by the power shelf **12** is overvoltage, overcurrent, and overloaded, may be determined.

[0019] The female cable **14** is electrically connected to the load **15**. The female cable **14** is configured to transfer the electrical signals outputted by the power shelf **12** to the load **15**, for powering on the load **15** by the power shelf **12**. The load **15** may be an electrical device of electrical consuming, such as servers, distribution panels. The number of the load **15** may be multiple, the multiple of the loads **15** are electrically connected to the female cable **14**, for receiving a power supply voltage and a power supply current provided by the power shelf **12**.

[0020] The power supply detection apparatus **13** may include several structures. Description will be made with reference to the power supply detection apparatus **13a** shown in FIG. 2, and the power supply detection apparatus **13b** as shown in FIG. 3 as examples. The present application does not limit the structure type of the power supply detection apparatus **13**.

[0021] Referring to FIG. 2, FIG. 2 shows a first embodiment of a structure of the power supply detection apparatus **13a** provided by the present application. The power supply detection apparatus **13a** includes a power shelf connector **131a**, a positive copper wire **132a**, a negative copper wire **133a**, a female cable connector **134a**, and a transformer **135a**.

[0022] The power shelf connector **131a** is configured to electrically connect with an output terminal of the power shelf **12**. The power shelf connector **131a** is made of metal material, and is in an I-shape. The power shelf connector **131a** includes a first output portion **131a_1** and a second output portion **131a_2**. The first output portion **131a_1** is electrically connected to one of the positive copper wire **132a** and the negative copper wire **133a**, and the second output portion **131a_2** is electrically connected to another of the positive copper wire **132a** and the negative copper wire **133a**.

[0023] In some embodiments, each of the first output portion **131a_1** and the second output portion **131a_2** define at least one through hole, a terminal of each of the positive copper wire **132a** and the negative copper wire **133a** connected to the first output portion **131a_1** or the second output portion **131a_2** is set with a fastener, such as a bolt, a nut, a screw, a washer, a rivet, and the like, thus the terminal of each of the positive copper wire **132a** and the negative copper wire **133a** is

connected to the power shelf connector **131a** through the at least one through hole of the positive copper wire **132a** and the negative copper wire **133a**.

[0024] The terminal of each of the positive copper wire **132a** and the negative copper wire **133a** is connected to the first output portion **131a_1** or the second output portion **131a_2** through the corresponding faster, another terminal of each of the positive copper wire **132a** and the negative copper wire **133a** is bundled in the female cable connector **134a**. For matching with a larger output current of the power shelf **12**, both of the positive copper wire **132a** and the negative copper wire **133a** are set to include a copper bundle with multiple copper wires.

[0025] The female cable connector **134a** is configured to connect with the female cable **14**. The female cable connector **134a** may be a clip connector, a bus bar connector, and the like.

[0026] The transformer **135a** extends through the positive copper wire **132a** or the negative copper wire **133a**. The transformer **135a** is configured to measure a current passing through the positive copper wire **132a** or the negative copper wire **133a** and/or a voltage passing through the positive copper wire **132a** or the negative copper wire **133a**, for obtaining outputting parameters of the power shelf **12**. The transformer **135a** may be a current transformer, or a voltage transformer, and the like. Therefore, there is no need to set with a current/voltage sensor being connected to the positive copper wire **132a** or the negative copper wire **133a** for obtaining the corresponding current or the voltage. The circuit structure of the power supply detection apparatus **13a** is simplified, and a failure rate of the power supply detection apparatus **13a** is decreased.

[0027] Referring to FIG. 3, FIG. 3 shows a second embodiment of the power supply detection apparatus **13b** provided by the present application. The power supply detection apparatus **13b** includes a power shelf connector **131b**, a positive copper bar **132b**, a negative copper bar **133b**, a female cable connector **134b**, a transformer **135b**, and a casing **136**. All of the power shelf connector **131b**, the positive copper bar **132b**, the negative copper bar **133b**, the female cable connector **134b**, and the transformer **135b** are received in the casing **136**.

[0028] The power shelf connector **131b** includes a first connection portion **131b_1** and a second connection portion **131b_2**. Both of the first output portion **131b_1** and the second output portion **131b_2** are made of metal material. The first connection portion **131b_1** is electrically connected to the positive copper bar **132b**, and the second connection portion **131b_2** is electrically connected to the negative copper bar **133b**.

[0029] In some embodiments, each of the first connection portion **131b_1** and the second connection portion **131b_2** define at least one through hole, a terminal of each of the positive copper bar **132b** and the negative copper bar **133b** connected to the first connection portion **131b_1** or the second connection portion **131b_2** is set with a fastener, such as a bolt, a nut, a screw, a washer, a rivet, and the like, thus the terminal of each of the positive copper bar **132b** and the negative copper bar **133b** is connected to the power shelf connector **131b** through the at least one through hole of the positive copper bar **132b** and the negative copper bar **133b**.

[0030] The positive copper bar **132b** includes a first bent component **132b_1** and a second bent component **132b_2**. The first bent component **132b_1** includes a first bent portion **132b_11**, a second bent portion **132b_12**, and a third bent portion **132b_13**. The second bent component **132b_2** includes a fourth bent portion **132b_21**, a fifth bent portion **132b_22**, and a sixth bent portion **132b_23**.

[0031] A surface of the first bent portion **132b_11** is connected to the first connection portion **131b_1** of the power shelf connector **131b**. Another surface of the first bent portion **132b_11** is perpendicularly connected to a sidewall of the second bent portion **132b_12**. Another opposite sidewall of the second bent portion **132b_12** is perpendicularly connected to a first sidewall of the third bent portion **132b_13**. That is, the first bent portion **132b_11** is parallel to the third bent portion **132b_13**.

[0032] The third bent portion **132b_13** is integrally formed with the fourth bent portion **132b_21**. The first sidewall of the third bent portion **132b_13** is a first sidewall of the fourth bent portion

132b_21. A second sidewall of the fourth bent portion **132b_21** is a sidewall opposite to the first sidewall. The second sidewall of the fourth bent portion **132b_22** is perpendicularly connected to a sidewall of the fifth bent portion **132b_22**. That is, the fifth bent portion **132b_22** is parallel with the second bent portion **132b_12**.

[0033] An opposite sidewall of the fifth bent portion **132b_22** is perpendicularly connected to the sixth bent portion **132b_23**. The sixth bent portion **132b_23** is connected to a positive electrode terminal of the female cable connector **134b**. A negative electrode terminal of the female cable connector **134b** is electrically connected to the second connection portion **131b_2** of the power shelf connector **131b** through the negative copper bar **133b**.

[0034] Therefore, by comparing with the power supply detection apparatus **13a** with the positive copper wire **132a** and the negative copper wire **133a** in FIG. 2, the power supply detection apparatus **13b** in FIG. 3 uses the positive copper bar **132b** and the negative copper bar **133b**, a surface area of a conductor is obviously increased, therefore, it is more easier to match with a larger output current of the power shelf **12**, and a cost of wire cables is saved.

[0035] Because of the second bent portion **132b_12** being parallel with the fifth bent portion **132b_22**, and the second bent portion **132b_12** being connected to the fifth bent portion **132b_22** through the third bent portion **132b_13** and the fourth bent portion **132b_21**, the second bent portion **132b_12**, the third bent portion **132b_23**, the fourth bent portion **132b_21**, and the fifth bent portion **132b_22** form a cavity.

[0036] The transformer **135b** extends through the cavity formed by the second bent portion **132b_12**, the third bent portion **132b_23**, the fourth bent portion **132b_21**, and the fifth bent portion **132b_22**. Therefore, the positive copper bar **132b** includes multiple bent portions, for electrically connecting the power shelf connector **131b** with the female cable connector **134b**. By comparing with the power supply detection apparatus **13a** with the positive copper wire **132a** and the negative copper wire **133a** in FIG. 2, a height of each of the positive copper bar **132b** and the negative copper bar **133b** in the power supply detection apparatus **13b** in FIG. 3 is reduced, a sum height of the transformer **135b** and the positive copper bar **132b** is not decided by a height of the copper bundle added with a height of the transformer **135a**, but is decided by a height of the third bent portion **132b_13**. Therefore, a height of the casing **136** may be set to be lower, for decreasing a sum height and a volume of the power source detection apparatus **13b**, and the power source detection apparatus **13b** may be matched with the cabinet **10** with a lower height, thus a adaptability of the power source detection apparatus **13b** is improved.

[0037] In addition, because of the power source detection apparatus **13b** with the casing **136** is an integrative apparatus. By comparing with the power source detection apparatus **13a**, the power shelf connector **131b** in the power source detection apparatus **13b** is not manually connected to the power shelf **12**, and then the positive copper wire **132a** and the negative copper wire **133b** are manually connected to the power shelf connector **131b**. Therefore, the assembly and operation steps are simplified, and an assembly efficiency of the power source detection apparatus **13b** is improved.

[0038] The structure of the female cable connector **134b** is the same as the female cable connector **134a**, the details will not be described again.

[0039] Referring to FIG. 4, FIG. 4 shows the power supply detection apparatus **13b** of FIG. 3 from another perspective provided by the present application.

[0040] In some embodiments, the casing **136** further includes a first output terminal **1361** and a second output terminal **1362**. The first output terminal **1361** and the second output terminal **1362** are electrically connected to the transformer **135b**. The first output terminal **1361** is an output terminal of a voltage signal of the transformer **135b**, and the second output terminal **1362** is an output terminal of a current signal of the transformer **135b**. The first output terminal **1361** and the second output terminal **1362** may be connected to an oscilloscope. Waveforms of the voltage signal and the current signal are displayed on the oscilloscope. Thus, the parameters of the power shelf **12**

are obtained, such as the output voltage, the output current, and the output power, and the like. [0041] In some embodiments, the casing **136** also includes a voltage input terminal **1363**. The voltage input terminal **1363** is also connected to the transformer **135b**. The voltage input terminal **1363** may be connected to an external power supply. The external power supply may provide a power supply to the transformer **135b** through the voltage input terminal **1363**.

[0042] Therefore, the power supply detection apparatus **13** and the cabinet **10** provided by the present application, may be conveniently and quickly detect electrical parameters, such as a current and a voltage outputted by the power shelf **12**.

[0043] Those skilled in the art will recognize that the above described embodiments are only intended to illustrate the invention and are not intended to limit the invention, and numerous possible modifications and variations within the spirit of the invention will fall within the scope of the invention.

Claims

1. A power supply detection apparatus comprises: a casing; a power shelf connector electrically connected to a power shelf and configured to receive a power supply from the power shelf; a cooper bar fixedly connected to the power shelf connector, and comprising a first bent component and a second bent component; the first bent component and the second component bent relative to each other to form a cavity; and a transformer extending through the cavity and configured to detect an output current and/or an output voltage of the power shelf.
2. The power supply detection apparatus of claim 1, wherein the first bent component comprises a first bent portion, a second bent portion, and a third bent portion; a surface of the first bent portion is connected to the first connection portion of the power shelf connector; another surface of the first bent portion is perpendicularly connected to a sidewall of the second bent portion; a sidewall of the second bent portion, opposite to the sidewall of the second bent portion connected to the first bent portion, is perpendicularly connected to a first sidewall of the third bent portion; the first bent portion is parallel to the third bent portion.
3. The power supply detection apparatus of claim 2, wherein the second bent component comprises a fourth bent portion and a fifth bent portion; the third bent portion is integrally formed with the fourth bent portion; a sidewall of the fourth bent portion, opposite to the first sidewall of the third bent portion, is perpendicularly connected to a first sidewall of the fifth bent portion; the fifth bent portion is parallel with the second bent portion.
4. The power supply detection apparatus of claim 3, wherein the cavity is further defined between the second bent portion, the third bent portion, the fourth bent portion, and the fifth bent portion.
5. The power supply detection apparatus of claim 3, wherein the second bent component further comprises a sixth bent portion; the sixth bent portion is perpendicularly connected to a second sidewall of the fifth bent portion opposite to the first sidewall of the fifth bent portion.
6. The power supply detection apparatus of claim 5, wherein the power supply detection apparatus further comprises a female cable connector; the female cable connector is configured to be connected with a female cable; the female cable connector is fixedly connected to the sixth bent portion.
7. The power supply detection apparatus of claim 1, wherein the power shelf connector defines at least one through hole; the cooper bar is fixedly connected to the power shelf connector with a fastener through the at least one through hole.
8. The power supply detection apparatus of claim 1, wherein the casing further comprises a first output terminal and a second output terminal; the first output terminal and the second output terminal are electrically connected to an oscilloscope; the transformer outputs a voltage signal through the first output terminal, and/or output a current signal through the second output terminal to indicate respectively an output voltage and/or an output current of the power shelf.

- 9.** The power supply detection apparatus of claim 1, wherein the casing further comprises a voltage input terminal; the voltage input terminal is connected between with the transformer and an external power supply; the voltage input terminal transmits a power supply provided by the external power supply to the transformer.
- 10.** A cabinet comprises: a power shelf, a female cable; and a power supply detection apparatus comprising: a casing; a power shelf connector electrically connected to a power shelf and configured to receive a power supply from the power shelf; a cooper bar fixedly connected to the power shelf connector, and comprising a first bent component and a second bent component; the first bent component and the second component bent relative to each other to form a cavity; and a transformer extending through the cavity and configured to detect an output current and/or an output voltage of the power shelf.
- 11.** The cabinet of claim 10, wherein the first bent component comprises a first bent portion, a second bent portion, and a third bent portion; a surface of the first bent portion is connected to the first connection portion of the power shelf connector; another surface of the first bent portion is perpendicularly connected to a sidewall of the second bent portion; a sidewall of the second bent portion, opposite to the sidewall of the second bent portion connected to the first bent portion, is perpendicularly connected to a first sidewall of the third bent portion; the first bent portion is parallel to the third bent portion.
- 12.** The cabinet of claim 11, wherein the second bent component comprises a fourth bent portion and a fifth bent portion; the third bent portion is integrally formed with the fourth bent portion; a sidewall of the fourth bent portion, opposite to the first sidewall of the third bent portion, is perpendicularly connected to a first sidewall of the fifth bent portion; the fifth bent portion is parallel with the second bent portion.
- 13.** The cabinet of claim 12, wherein the second bent portion, the third bent portion, the fourth bent portion, and the fifth bent portion form the cavity.
- 14.** The cabinet of claim 12, wherein the second bent component further comprises a sixth bent portion; the sixth bent portion is perpendicularly connected to a second sidewall of the fifth bent portion opposite to the first sidewall of the fifth bent portion.
- 15.** The cabinet of claim 14, wherein the power supply detection apparatus further comprises a female cable connector; the female cable connector is configured to connect with a female cable; the female cable connector is fixedly connected to the sixth bent portion.
- 16.** The cabinet of claim 10, wherein the power shelf connector defines at least one through hole; the cooper bar is fixedly connected to the power shelf connector with a fastener through the at least one through hole.
- 17.** The cabinet of claim 10, wherein the casing further comprises a first output terminal and a second output terminal; the first output terminal and the second output terminal are electrically connected to an oscilloscope; the transformer outputs a voltage signal through the first output terminal, and/or output a current signal through the second output terminal; the voltage signal and the current signal are configured to indicate an output voltage and/or an output current of the power shelf.
- 18.** The cabinet of claim 10, wherein the casing further comprises a voltage input terminal; the voltage input terminal is connected between with the transformer and an external power supply; the voltage input terminal transmits a power supply provided by the external power supply to the transformer.
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